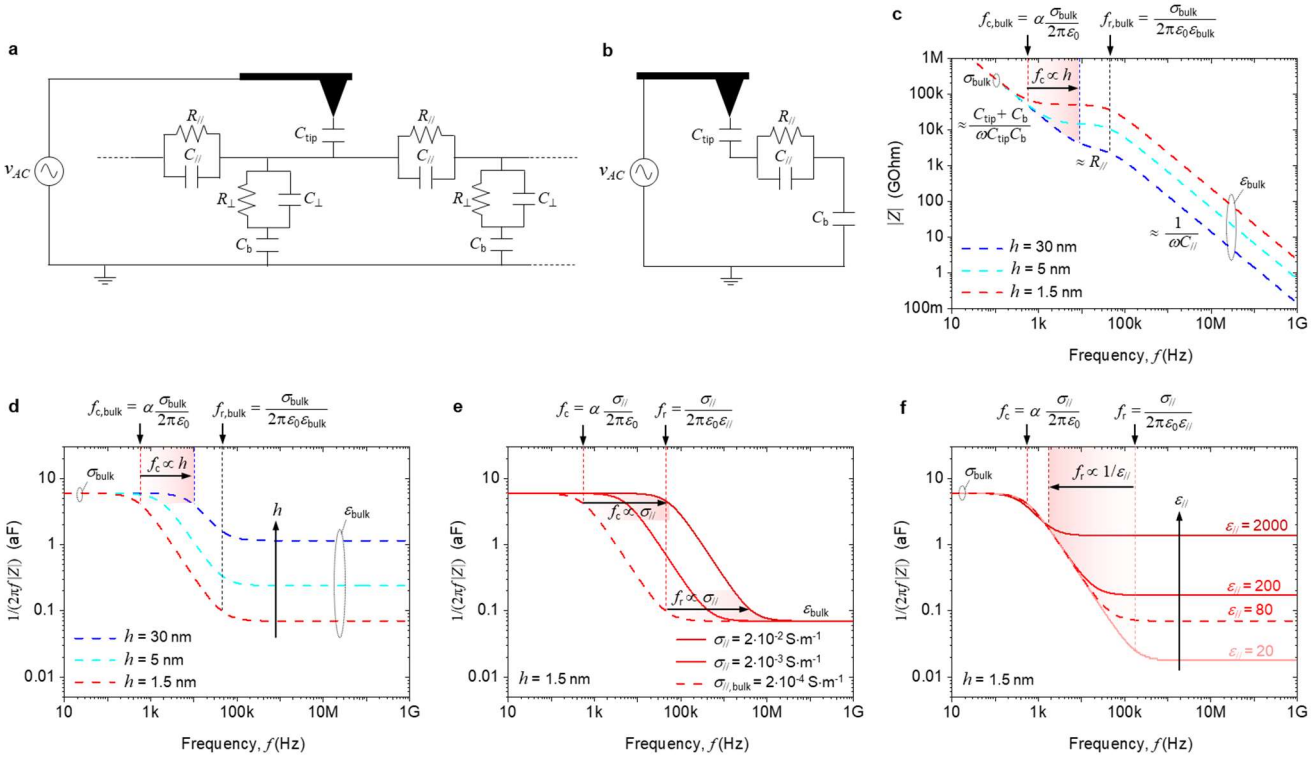
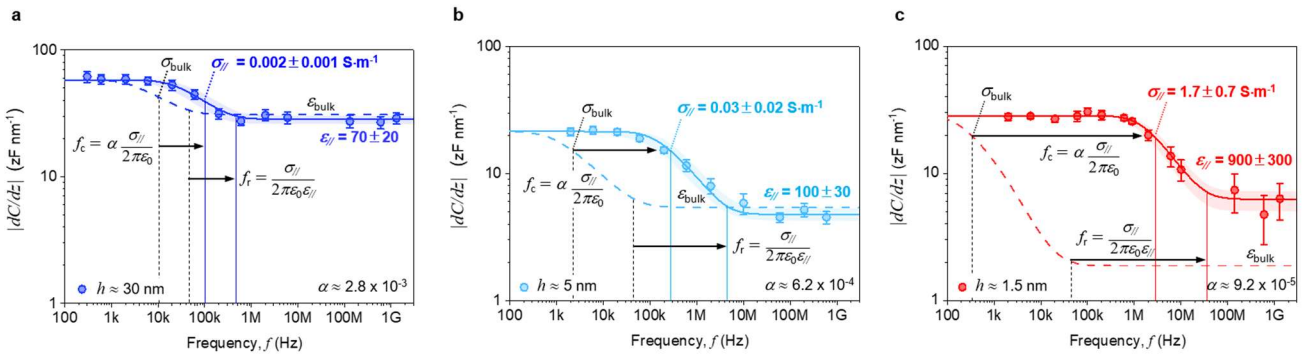


Extended Data Fig. 9 | Simulated dielectric spectra using equivalent electrical circuits. (**a**) Approximate impedance model for our measurement geometry using a distributed RC network that includes both in-plane and out-of-plane elements to model the water channel. (**b**) Simplified in-plane equivalent circuit that takes into account the negligible contribution of the out-of-plane impedance. (**c**) Calculated impedance using different channel thicknesses (h = 30, 5 and 1.5 nm) and bulk water's properties. (**d**) Same as in (c) but plotting the effective capacitance, $1/(|Z|2\pi f)$. (**e**) Same as (d) for h = 1.5 nm using different $\sigma_{//}$. (**f**) Same as (e) but for different $\varepsilon_{//}$. Other parameters used in simulations (c-f): R = 100 nm, θ = 25°, z_{scan} = 20 nm, l^* = 3 µm, w_c = 200 nm, H_{top} = 50 nm and H = 100 nm.



Extended Data Fig. 10| Fitting experimental spectra. (**a-c**) Symbols are the experimental data of Fig. 2a of the main text before normalization. Bars are the random noise level as measured at each f. The solid curves are the best fits using the 3D numerical model with water's $\varepsilon_{//}$ and $\sigma_{//}$ as the only fitting parameters (labels). The light-shaded regions indicate the uncertainty in the extracted values of $\varepsilon_{//}$ and $\sigma_{//}$. Dashed curves: 3D numerical simulations for the same devices assuming bulk water's properties ($\varepsilon_{//} = \varepsilon_{\perp} = \varepsilon_{bulk} = 80$, $\sigma_{//} = \sigma_{\perp} = \sigma_{bulk} = 2 \times 10^{-4}$ S·m^{-1}). The vertical lines indicate calculated values of f_c and f_r for the case of bulk water (black dashed) and the experimentally extracted water parameters (colored solid). To evaluate the frequencies, we used Eqs. (5-6) and α obtained from the full 3D numerical simulations: (**a**) $\alpha = 2.8 \times 10^{-3}$, (**b**) $\alpha = 6.2 \times 10^{-4}$ and (**c**) $\alpha = 9.2 \times 10^{-5}$. $\varepsilon_{\perp}$ is set to the values measured in Ref.[22]. Other experimental parameters used in the simulations: (**a**) h = 30 nm, $\varepsilon_{\perp}$ = 32, $\sigma_{\perp} = 2 \cdot 10^{-4}$ S·m^{-1}, R = 84 nm, θ = 26°, H_{top} = 21 nm, H = 53 nm, w = 210 nm, z_{scan} = 19 nm; (**b**) h = 5 nm, $\varepsilon_{\perp}$ = 6.5, $\sigma_{\perp} = 2 \cdot 10^{-4}$ S·m^{-1}, R = 62 nm, θ = 17°, H_{top} = 34 nm, H = 45 nm; w = 220 nm, z_{scan} = 18 nm; (**c**) h = 1.5 nm, $\varepsilon_{\perp}$ = 2.1, $\sigma_{\perp} = 2 \cdot 10^{-4}$ S·m^{-1}, R = 91 nm, θ = 27°, H_{top} = 50 nm, H = 76 nm, w = 210 nm, z_{scan} = 19 nm.